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Methods to detect pathological shortening or elongation of circular DNA in human cells

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Abstract

Circular DNA molecules, including mitochondrial DNA (mtDNA) and extrachromosomal circular DNA (ecDNA), are critical genetic elements in human cells. Pathological alterations in their length—through deletions, duplications, or complex rearrangements—are directly implicated in a spectrum of diseases, from inherited mitochondrial disorders to cancer progression and therapy resistance. This review provides a comprehensive analysis of the methodologies available for detecting these specific structural changes. We systematically evaluate classical techniques, such as long-range PCR, Southern blotting, and electron microscopy, which offer direct visualization and validation. Furthermore, we examine modern high-throughput approaches, including short- and long-read sequencing technologies that enable genome-wide discovery and precise breakpoint mapping at single-nucleotide resolution. Special emphasis is placed on quantitative methods like digital droplet PCR (ddPCR) for ultra-sensitive detection and monitoring in clinical settings. A comparative framework is presented to guide the selection of appropriate methods based on resolution, throughput, quantifiability, and cost. We conclude that an integrated, multi-method strategy is often essential for robust analysis, from initial discovery to clinical validation. The accurate detection of circular DNA length alterations is rapidly evolving from a research tool into a cornerstone of precision diagnostics, offering significant potential for advancing our understanding and management of complex human diseases.

Keywords: Mitochondrial DNA, Extrachromosomal Circular DNA, Structural Variation, Next-Generation Sequencing, Digital PCR, Molecular Diagnostics.

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Introduction

The human genome is predominantly linear, housed within chromosomes in the nucleus. However, a crucial and dynamic fraction of genetic material exists in a circular conformation. This circular DNA (circDNA) plays vital roles in cellular physiology, and aberrations in its structure, particularly changes in its length, are directly linked to severe pathologies, ranging from metabolic disorders to cancer. Detecting these alterations—pathological shortening or elongation—is therefore a critical challenge in molecular diagnostics and research. This review focuses on systematically analyzing the methodologies available for this specific purpose, evaluating their principles, applications, and limitations in the context of human cellular pathology.

Biological Significance of Circular DNA in Human Cells

Circular DNA in human cells primarily exists in two major forms: mitochondrial DNA and extrachromosomal circular DNA, each with distinct origins and functions.

Mitochondrial DNA (mtDNA) is a hallmark of endosymbiotic origin. This small (approximately 16.6 kbp), double-stranded, circular genome is present in hundreds to thousands of copies per cell, a state known as polyploidy, and is strictly maternally inherited (Anderson et al., 1981). It encodes 37 genes essential for oxidative phosphorylation (OXPHOS), the process that generates most of the cell's adenosine triphosphate (ATP). Unlike nuclear DNA, mtDNA lacks protective histones and has a limited repair capacity, making it particularly susceptible to damage. Pathological events often lead to large-scale deletions or, less frequently, duplications. The so-called "common deletion" of 4977 bp, for instance, removes several key genes and is a well-documented biomarker in conditions like Kearns-Sayre syndrome and sporadic inclusion body myositis, and is also associated with aging and Parkinson's disease (Kraytsberg et al., 2006). Such deletions effectively shorten the mtDNA molecule, while duplications lead to its elongation, both disrupting the stoichiometry and function of the OXPHOS complexes.

Extrachromosomal circular DNA (ecDNA) represents a broad category of circular DNAs derived from the nuclear genome. Once considered rare, they are now recognized as major drivers of genomic plasticity, especially in cancer. Oncogenic ecDNA are amplifications of genomic regions harboring powerful driver genes, such as MYCN in neuroblastoma or EGFR in glioblastoma (Verhaak et al., 2019). These circular amplicons lack centromeres and segregate randomly during cell division, leading to rapid intra-tumoral heterogeneity and adaptation. Crucially, their size and structure are not static; they can undergo rearrangement, recombination, and changes in length (e.g., through head-to-tail duplications forming larger circles or multimers) during tumor evolution, contributing to drug resistance and aggressiveness (Wu et al., 2019). Beyond oncogenes, ecDNA also includes smaller, non-coding circles (eccDNA) whose functions are still being elucidated, as well as episomes like those of certain viruses.

Pathological Consequences of Altered Length

The physical length of a circDNA molecule is intrinsically tied to its genetic content and structural integrity. Alterations trigger cascading pathological effects.

Shortening via Deletions: The primary consequence is the loss of genetic material. In mtDNA, deletions remove essential tRNA, rRNA, or protein-coding genes. This creates a state of heteroplasmy—a mix of wild-type and deleted molecules within a cell. When the proportion of deleted molecules exceeds a tissue-specific threshold, it results in a bioenergetic crisis. The loss of OXPHOS subunits compromises ATP production, increases reactive oxygen species (ROS) generation, and ultimately leads to mitochondrial cytopathies. These disorders often affect high-energy-demand tissues like muscle, brain, and nerves, manifesting as myopathies, encephalopathies, and neuropathies. Furthermore, the accumulation of deletion-bearing mtDNA is a hallmark of aging in post-mitotic tissues.

Elongation via Duplications and Insertions: Elongation often implies an increase in gene copy number (dosage) on a single molecule. For ecDNA carrying oncogenes, this can lead to massive overexpression, as the gene is no longer constrained by the regulatory context of its native chromosomal locus (e.g., topologically associating domains). A single elongated ecDNA molecule can carry dozens of copies of an oncogene, creating "transcription hubs" that dramatically elevate mRNA output (Hung et al., 2021). Structurally, elongated or multimeric circles may be more unstable, prone to further recombination, and may facilitate the co-amplification of adjacent regulatory elements or resistance genes. This genomic instability and enhanced oncogene expression are strongly associated with tumor progression, therapeutic failure, and poor prognosis across multiple cancer types. The formula for calculating the approximate copy number of a gene on an ecDNA, while simplistic, illustrates the concept: Gene Copies per Cell ≈ (Number of ecDNA molecules per cell) × (Number of gene copies per ecDNA molecule). This multiplicative effect underscores the potency of elongated ecDNA.

Aim of the Article

Given the profound clinical implications of circDNA length alterations, accurate detection is paramount. Numerous techniques exist for studying circDNA, but they differ vastly in their ability to resolve, quantify, and characterize changes in physical length versus simply confirming circularity or presence. Methods like standard PCR for back-splice junctions in circular RNAs, for example, do not inform on molecule size. This article aims to systematically categorize and evaluate methods specifically tailored for detecting pathological shortening or elongation of circular DNA molecules in human cells. We will focus on techniques that provide direct or inferential evidence of size change, from classical molecular biology approaches to next-generation sequencing and cutting-edge microscopy. The goal is to provide a clear framework for researchers and clinicians to select the most appropriate methodological arsenal based on their specific question, whether it is screening for known mtDNA deletions, validating a novel ecDNA amplicon, or discovering entirely new classes of size-altered circular DNA in disease.

Electrophoresis and PCR-Based Detection Methods

Before the advent of high-throughput genomics, the detection of structural variations in DNA relied heavily on techniques combining enzymatic amplification, restriction digestion, and electrophoretic separation. These methods remain foundational, providing a direct, visual, and often quantitative assessment of DNA length, forming the first line of investigation for circular DNA alterations.

Long-Range Polymerase Chain Reaction (LR-PCR)

Principle: Long-range PCR is an adaptation of the standard PCR protocol optimized to amplify significantly longer DNA fragments, typically spanning several kilobase pairs. This is achieved using thermostable DNA polymerases (or enzyme blends) with high processivity and proofreading activity, coupled with optimized buffer systems. To detect deletions or duplications in circular DNA, primers are designed to flank the suspected alteration. For a deletion, the primer binding sites are brought closer together on the mutant molecule, resulting in a shorter amplicon compared to the full-length wild-type product. Conversely, a duplication event leads to a proportionally longer PCR product. The size difference is then resolved by standard agarose gel electrophoresis.

Application: LR-PCR has been a workhorse for screening known, large-scale deletions in mitochondrial DNA. The classic 4977 bp "common deletion" in mtDNA is a prime example, routinely detected by amplifying across the breakpoint to yield a distinct, smaller band visible on a gel (Kraytsberg et al., 2006). This method is effective for surveying known "hotspot" deletions in patient samples, such as in muscle biopsies from individuals with mitochondrial myopathies. While less common, it can also hint at duplications if a larger-than-expected band is observed, although this requires careful optimization to rule out non-specific amplification. The sensitivity of LR-PCR allows for the detection of heteroplasmy levels down to approximately 1-5%, depending on the specific deletion and tissue type.

Limitations: The primary limitations of LR-PCR are its targeted nature and semi-quantitative output. It is a hypothesis-driven technique, requiring prior knowledge of the approximate breakpoint regions to design primers. It does not provide the exact nucleotide coordinates of the deletion or duplication junction. Quantification of heteroplasmy is imprecise, typically estimated from band intensity on a gel, which is non-linear and sensitive to technical variables. Furthermore, the high GC-content and complex secondary structure of mtDNA can lead to amplification artifacts, stalling, or preferential amplification of smaller (deleted) molecules, potentially skewing the observed heteroplasmy ratio. Finally, its utility for analyzing complex ecDNA populations is limited due to their variable and often unknown sequences.

Southern Blot Analysis

Principle: Southern blotting remains a definitive, gold-standard technique for validating large DNA rearrangements. The process involves digesting total genomic DNA (containing both nuclear and circular DNA) with one or more restriction enzymes that have known cutting sites in the target sequence. The digested fragments are separated by size using agarose gel electrophoresis. The DNA in the gel is then denatured and transferred (blotted) onto a solid membrane, where it is immobilized. A labeled (radioactive or non-radioactive) DNA probe complementary to a specific sequence within the target circular DNA is hybridized to the membrane. After washing, the probe binds only to its complementary fragments, which are then visualized, revealing their sizes.

Application: For circular DNA analysis, Southern blot is unparalleled in its ability to distinguish topological forms. When analyzing mtDNA, digestion with an enzyme that cuts once within the genome linearizes all circular mtDNA molecules to a uniform full-length fragment. The presence of a deletion is indicated by an additional, shorter hybridizing band. The intensity of this band relative to

the full-length band provides a robust quantitative measure of heteroplasmy. Critically, by using enzymes that do not cut within the circle, one can distinguish supercoiled, relaxed circular, and linear forms based on their differential migration, confirming the circular nature of the DNA (Shadel & Clayton, 1997). For ecDNA, Southern blot can confirm the circular structure of an amplicon and estimate its size, distinguishing it from integrated, linear amplifications.

Limitations: The major drawbacks of Southern blotting are its labor-intensive and time-consuming protocol, low throughput, and requirement for substantial amounts (micrograms) of high-quality, high-molecular-weight DNA. It is poorly suited for screening large numbers of samples or for detecting small indels (e.g., < 50 bp), where the size difference may be difficult to resolve on a standard agarose gel. The use of radioactive probes, while highly sensitive, involves safety and regulatory hurdles. Despite these limitations, its role as an orthogonal validation method for findings from PCR or sequencing is irreplaceable.

Two-Dimensional Gel Electrophoresis (2D-GE)

Principle: Two-dimensional gel electrophoresis separates DNA molecules based on two independent properties in two sequential electrophoresis steps. In the first dimension, DNA is separated by molecular weight (size) in a low-concentration agarose gel under conditions that minimize the effect of topology. A lane from this first-dimension gel is then excised, rotated 90 degrees, and embedded into a second agarose gel containing an intercalating dye like chloroquine or ethidium bromide. The second-dimension electrophoresis is run with the dye present, which alters the supercoiling density of DNA. Molecules are now separated based on their topology (e.g., supercoiled, relaxed circle, linear) at a given mass.

Application: 2D-GE is a powerful tool for analyzing the complex topology and size heterogeneity of circular DNA populations. It can resolve a mixture of circular DNA molecules of different sizes and different supercoiling states into a characteristic arc pattern. For instance, it can separate nicked (relaxed) circular mtDNA from supercoiled mtDNA and from linear fragments, providing insights into DNA damage and repair. In the study of ecDNA, 2D-GE has been used to demonstrate the circular nature of oncogenic amplicons and to reveal populations of circles of various sizes, including complex forms like catenanes or replication intermediates (Cohen, Lavi, et al., 1997). This method is uniquely capable of providing a "topological snapshot" of the circular DNA pool in a cell.

Limitations: The technique is notoriously technically challenging, requiring extensive optimization of running conditions, dye concentrations, and gel concentrations. It is low-throughput, not quantitative in a straightforward manner, and demands significant expertise to interpret the complex spot and arc patterns. Its resolution for precise size determination is inferior to direct sequencing or even Southern blotting. Consequently, 2D-GE is primarily a specialized research tool for fundamental studies of DNA structure and replication rather than a routine diagnostic or screening method for length alterations.

In summary, electrophoresis and PCR-based methods offer a spectrum of approaches from rapid screening (LR-PCR) to definitive validation (Southern blot) and topological analysis (2D-GE). Their common strength is the direct physical separation of molecules by size, providing intuitive visual evidence of length alterations. However, their reliance on prior sequence knowledge, limited

resolution for small changes, and low throughput necessitate complementary approaches for comprehensive analysis, as discussed in the following sections.

Microscopy-Based Detection Methods

Moving beyond indirect inference from fragment size, microscopy techniques provide a direct visual assessment of individual circular DNA molecules within their cellular context or in purified form. These methods transform abstract genetic data into tangible images, allowing for the unambiguous confirmation of circularity, precise physical length measurement, and observation of higher-order structural features. While often lower in throughput, they offer unparalleled validation and unique insights.

Transmission Electron Microscopy (TEM)

Principle: Transmission Electron Microscopy represents the most direct method for visualizing DNA molecules at the nanoscale. The technique involves preparing a purified DNA sample, often through gradient centrifugation or column-based isolation of circular DNA fractions. The DNA is then mounted on a grid, stained with heavy metal salts (typically uranyl acetate or platinum-based shadows), and dried. In the electron microscope, a beam of electrons is transmitted through the specimen. Areas where heavy metals have bound to the DNA (primarily to the phosphate backbone) scatter electrons more strongly, creating a dark, high-contrast image of the molecule against a brighter background. Individual molecules appear as fine threads, and their topology—linear, relaxed circular, or supercoiled—is immediately apparent.

Application: TEM's primary strength in circular DNA analysis is the provision of absolute, physical length measurements. By including an internal length standard (such as a DNA molecule of known size, e.g., bacteriophage phiX174 virion DNA at 5386 base pairs), the contour length of an unknown circle can be directly calculated from the micrograph. The length in micrometers is converted to base pairs using the established conversion factor of approximately 0.34 nm (or 3.4 Å) per base pair for B-form DNA. Therefore, the size in kilobase pairs (kbp) can be estimated as: Length (kbp) ≈ [Contour Length (μm) / 0.34] / 1000. This allows for the direct quantification of deletions or duplications. Historically, TEM was instrumental in the initial discovery and characterization of extrachromosomal circular DNA in cancer cells, such as double-minute chromosomes (Cox et al., 1965). Beyond mere length, TEM can reveal intricate structural details like branched structures (indicative of recombination or replication intermediates), "lariats," or the presence of bound proteins, which are invisible to sequencing-based methods. It serves as the ultimate orthogonal validation for circular structures proposed by other techniques.

Limitations: The utility of TEM is counterbalanced by significant practical constraints. The sample preparation is demanding, requiring highly purified DNA free of contaminants like proteins, RNA, and salts that can obscure the image. The process is low-throughput, labor-intensive, and expensive, reliant on specialized equipment and expertise. Image acquisition and analysis are time-consuming and can be subjective, especially for complex or overlapping molecules. Statistical analysis of length distributions requires measuring hundreds of molecules, which is not feasible for routine

diagnostics. Furthermore, TEM provides a static snapshot of purified DNA, divorced from its native nuclear or mitochondrial context, limiting insights into cellular dynamics and copy number.

Fluorescence In Situ Hybridization (FISH) and Its Derivatives

Principle: Fluorescence in situ hybridization (FISH) bridges the gap between molecular analysis and cytology. It involves designing fluorescently labeled nucleic acid probes complementary to a target sequence within the circular DNA. These probes are hybridized to fixed cells or chromosomes on a microscope slide. After washing away unbound probe, the location and intensity of the fluorescent signal are visualized using a fluorescence microscope. This allows for the detection of DNA targets within the morphological context of the nucleus.

Application: The application of FISH to circular DNA analysis has evolved into two powerful, complementary variants:

- Classical Interphase FISH: This is the standard method for detecting ecDNA within the nuclei of tumor cells. Probes targeting specific oncogenes (e.g., MYCN, EGFR) produce bright fluorescent foci. The absence of these foci on chromosomes during metaphase, coupled with their presence in the nucleoplasm during interphase, is a classic cytogenetic signature of ecDNA (Hahn et al., 2021). While not a precise metric, the size and intensity of these foci can offer a qualitative indication: exceptionally large or bright foci may suggest the presence of concatenated or highly amplified circles, implying an increase in the local target sequence length per molecule. This technique has been pivotal in linking ecDNA presence with poor prognosis across cancer types.

- Fiber-FISH (DNA Fiber Fluorescence In Situ Hybridization): This technique dramatically increases resolution for length analysis. Cells are lysed on a slide, and the released DNA is stretched uniformly by gravity or molecular combing, transforming three-dimensional chromatin into linear, elongated fibers. Hybridization with multiple, differentially colored probes along a target region allows for the direct visualization of sequence order and spacing. For circular DNA analysis, Fiber-FISH is transformative. It can directly visualize and measure the physical length of a stretched circular molecule (now linearized on the fiber). By hybridizing probes that flank a suspected duplication, the increased distance between the probe signals provides a direct, visual, and quantifiable measure of the inserted sequence length. It is the method of choice for physically mapping the complex structure of large ecDNA amplicons, confirming tandem duplications, and estimating the number of gene repeats within a single circle (Shoshani et al., 2021). The length measurement on the fiber (in micrometers) can be converted to kilobase pairs using a calibration standard hybridized in parallel, applying the same physical constant (0.34 nm/bp) used in TEM.

Limitations: While powerful, FISH-based methods have inherent challenges. Classical FISH provides only qualitative or semi-quantitative data on size. Fiber-FISH, while quantitative, is technically difficult; the quality of fiber preparation and stretching consistency are critical and can vary, introducing measurement error. The stretching factor must be carefully calibrated for each experiment. Both methods require a priori knowledge of the target sequence to design specific probes, making them unsuitable for discovering novel, uncharacterized circular DNAs. The

throughput is low, and the analysis, particularly of Fiber-FISH images, requires specialized software and expertise to accurately measure signal distances and assign structures.

In conclusion, microscopy-based methods serve as crucial tools in the circular DNA analysis toolkit. TEM provides the definitive, high-resolution proof of circularity and direct physical measurement in a purified system. FISH and Fiber-FISH contextualize these molecules within the cell, enabling the correlation of structural features like length with cellular phenotypes and providing a powerful visual means to map and measure specific sequences on individual DNA fibers. Their role is often that of a confirmatory and descriptive technique, complementing the higher-throughput, sequence-centric approaches that will be discussed next.

Methods Based on High-Throughput Sequencing (NGS)

The advent of Next-Generation Sequencing (NGS) has fundamentally transformed the detection and analysis of circular DNA, shifting the paradigm from targeted interrogation to comprehensive, genome-wide discovery. NGS provides a digital, sequence-based readout that can identify novel alterations with single-nucleotide precision and quantify them across entire populations of molecules. The choice between short-read and long-read platforms dictates the resolution and scope of structural insights possible.

Short-Read Sequencing (Illumina)

Principle: Short-read sequencing (e.g., Illumina) generates millions of short DNA fragments (reads), typically 75-300 base pairs in length. Detection of circular DNA and its length alterations is performed indirectly through computational analysis of how these reads map (align) back to a reference genome. The circular nature and structural variants are inferred from specific, statistically significant anomalies in the mapping patterns.

Application: Two primary bioinformatic signatures are leveraged for detecting length changes in circular DNA:

1. Discordant Read Pairs and Split Reads: This is the key signal for identifying the precise breakpoints of deletions or duplications. In a standard Illumina library, two reads are sequenced from opposite ends of a DNA fragment. In a linear genome, they are expected to map to the reference in a specific inward-facing orientation and within a distance defined by the library's insert size. A circular junction, formed by a deletion or the head-to-tail ligation of an ecDNA, creates an aberrant genomic context. Reads spanning this junction will appear as "split reads," where one part of a read maps to one side of the junction and the remainder maps to a non-adjacent region. The corresponding mate pair will also map discordantly, with an unexpected orientation or distance. Algorithms specifically designed to detect circular DNA, such as CIRCexplorer2 or CIRI2, exploit these patterns to identify back-splice junctions in circular RNAs and structural breakpoints in other circDNAs (Zhang et al., 2016). For mtDNA, this approach can pinpoint the exact coordinates of deletions with high accuracy.

2. Read Depth (Coverage) Analysis: While less specific for circularity, coverage analysis is crucial for inferring duplications. A tandem duplication within a circular molecule will lead to a local increase in the sequencing read depth over the duplicated segment. The fold-increase in normalized coverage can approximate the copy number of the repeat within the circle. This is particularly useful for characterizing ecDNA amplicons, where steep, focal increases in coverage define the "amplicon structure" and highlight core amplified genes (Beroukhim et al., 2010). The ratio of coverage in the duplicated region versus a single-copy control region provides an estimate of the duplication multiplicity.

Limitations: The fundamental constraint of short-read sequencing is the physical length of the reads, which is far shorter than most circular DNA molecules or the repetitive elements within them. This leads to several critical shortcomings. First, it cannot fully resolve or "phase" long tandem repeats; a duplication of a 10 kbp segment cannot be assembled from 150 bp reads. Second, it cannot reliably distinguish between a true extrachromosomal circle and the same sequence integrated into a chromosome in a head-to-tail array, as both produce similar split-read and coverage signatures. Third, de novo assembly of complete circular sequences from short reads is computationally challenging and often ambiguous, especially for complex ecDNA. Finally, the need for PCR amplification during library preparation can introduce biases and obscure the natural topology of the DNA.

Long-Read Sequencing (Third-Generation: PacBio and Oxford Nanopore)

Principle: Long-read sequencing technologies from PacBio (Single Molecule, Real-Time or SMRT sequencing) and Oxford Nanopore Technologies (ONT) directly sequence single DNA molecules, producing reads that average tens of kilobases and can extend beyond 100 kbp. This leap in read length is revolutionary for circular DNA analysis, as a single read can now span an entire circular molecule or capture its complete junction structure in a single, uninterrupted data point.

Application: Long-read sequencing addresses the core limitations of short-read approaches:

1. Direct Junction Spanning and Precise Breakpoint Identification: A single long read can traverse the entire junction of a circular DNA molecule, capturing the exact nucleotide where the circle closes. For a deletion circle, this reveals both breakpoints in one read. For an ecDNA, it identifies the precise genomic coordinates fused to form the circle. This eliminates the need for indirect inference from read pairs, providing unambiguous evidence of circularity and the exact structure of the alteration (Zhao et al., 2021).

2. Complete Circular Molecule Sequencing: The entire mitochondrial genome (~16.6 kbp) can be sequenced in a single, continuous read on either platform. This allows for the de novo assembly of complete mtDNA genomes from individual reads, enabling the detection of complex heteroplasmy, multiple coexisting deletions, and precise haplotyping. For larger ecDNA, while a single read may not cover the entire circle (which can be hundreds of kbp), multiple overlapping long reads enable the complete, accurate, and unambiguous assembly of the circular contig, revealing its full sequence, gene order, and internal duplications without guesswork.

3. Epigenetic Analysis (Nanopore): A unique advantage of Oxford Nanopore sequencing is its ability to detect native base modifications, such as 5-methylcytosine (5mC), as the DNA strand passes through the pore. This allows for the simultaneous assessment of a circular DNA molecule's length, sequence, and methylation state in a single assay, providing a multi-modal view of its biology (Liu et al., 2021).

The power of long reads for structural analysis can be conceptualized by comparing the probability of spanning a junction. For a short read of length L_s (e.g., 150 bp) and a junction, the probability is negligible unless the read is centered exactly on the junction. For a long read of length L_l (e.g., 15,000 bp), the probability of spanning a junction within a molecule of size M is significantly higher and can be approximated as: $P_{span} \approx (L_l - J) / M$, where J is the minimal sequence needed to uniquely map across the junction (e.g., 50 bp). For mtDNA, this probability approaches 1.

Limitations: Historically, the trade-off for long read length was a higher per-base error rate (1-15% for raw reads) compared to Illumina's near-perfect accuracy. However, PacBio's HiFi mode, which uses circular consensus sequencing (CCS), achieves accuracy >99.9% by sequencing the same molecule multiple times, effectively mitigating this issue. For Nanopore, base-calling algorithms and improved chemistries have steadily reduced error rates. Other limitations include higher DNA input requirements, higher per-sample cost for deep coverage, and significantly larger and more complex data sets that demand substantial computational resources and specialized bioinformatic pipelines for assembly and variant calling. Despite these challenges, long-read sequencing is rapidly becoming the method of choice for definitive characterization of circular DNA length and structure.

In summary, NGS provides a powerful, scalable framework for circular DNA analysis. Short-read sequencing offers cost-effective discovery and quantification of alterations, particularly when leveraged with specialized bioinformatic tools. Long-read sequencing, however, delivers a paradigm shift by providing direct, unambiguous, and complete molecular resolution, making it an indispensable tool for validating discoveries and unraveling the complex architecture of pathological circular DNA.

Quantitative Methods for Clinical Diagnostics and Monitoring

While discovery and structural characterization are vital for research, the clinical management of diseases driven by circular DNA alterations demands robust, quantitative, and highly sensitive assays. These methods must provide precise measurements of mutation load (heteroplasmy for mtDNA or ecDNA prevalence in tumors) from often limited, suboptimal clinical samples like blood, biopsies, or liquid biopsies. Quantitative PCR (qPCR) and its advanced derivative, digital droplet PCR (ddPCR), have become the cornerstones of such applied diagnostic and monitoring workflows due to their speed, specificity, and quantitative power.

Quantitative Real-Time PCR (qPCR)

Principle: Quantitative real-time PCR measures the accumulation of fluorescent signal during each cycle of a polymerase chain reaction, allowing for the quantification of the initial amount of target

DNA. For detecting deletions or duplications in circular DNA, a comparative quantitation method (often the ΔΔCt method) is employed. This requires designing two sets of primers and probes:

1. A target assay that amplifies a sequence within the region suspected to be deleted or duplicated. For a common deletion, this assay would span the breakpoint junction, ensuring it only amplifies the mutant, shortened molecule.

2. A reference assay that amplifies a stable, single-copy region unaffected by the deletion. In mtDNA analysis, this is typically a region in the conserved, undeleted portion of the genome, such as the ND1 gene.

The relative quantity of the mutant DNA is determined by comparing its amplification efficiency (Ct value) to that of the reference DNA. The critical output is the heteroplasmy percentage, calculated as the proportion of mutant mtDNA molecules relative to the total (mutant + wild-type). A standard formula for estimating this from Ct values is: % Heteroplasmy = 100 / [1 + 2^(ΔCt)], where ΔCt = Ct(reference) - Ct(target). A ΔCt of 0 indicates 50% heteroplasmy (equal amounts), a negative ΔCt indicates >50%, and a positive ΔCt indicates <50%. For duplications, the logic is reversed; the target assay would amplify a junction indicative of the duplication, and an increased ratio relative to the reference suggests the presence of elongated molecules.

Application: qPCR is the workhorse for rapid, cost-effective screening of known, high-prevalence mutations. It is extensively used in diagnostic laboratories for quantifying common mtDNA deletions (like the 4977 bp "common deletion") in patients with suspected mitochondrial disorders (He et al., 2002). Its high throughput enables screening of large patient cohorts. Furthermore, it is indispensable for longitudinal monitoring. For example, the heteroplasmy level of a pathogenic mtDNA deletion in a patient's blood or urine sediment can be tracked over time to assess disease progression or, in rare cases, response to experimental therapies. In oncology, while less common for ecDNA due to sequence heterogeneity, qPCR can be used to track specific, known fusion junctions of ecDNA in cell-free DNA from liquid biopsies if the junction sequence is identified from a prior tumor biopsy.

Limitations: The primary limitation of qPCR is that it measures relative, not absolute, quantity. The result (% heteroplasmy) depends on the accuracy of the reference assay, which assumes the reference region is stably diploid (for nuclear DNA) or present at a constant copy number per genome (for mtDNA). This assumption can be violated; mtDNA copy number itself varies between tissues and under pathological conditions. The technique is also sensitive to inhibitors commonly found in clinical samples (e.g., heparin, heme) and requires high-quality DNA for reliable efficiency. Its dynamic range and sensitivity are ultimately constrained by the precision of Ct measurement, typically making reliable detection below 1-2% heteroplasmy challenging. Finally, it is entirely dependent on prior knowledge of the alteration for assay design.

Digital Droplet PCR (ddPCR)

Principle: Digital droplet PCR represents a paradigm shift in nucleic acid quantification by transforming the measurement from analog (based on reaction kinetics) to digital (based on binary counting). The reaction mixture—containing DNA template, primers, probes, and PCR reagents—is

partitioned into approximately 20,000 nanoliter-sized water-in-oil droplets. Each droplet effectively acts as an individual PCR micro-reactor. Following endpoint PCR amplification, each droplet is analyzed by a droplet reader. Droplets containing at least one copy of the target sequence will fluoresce positively (1), while those without the target remain negative (0). This binary readout allows for absolute quantification without the need for a standard curve. The concentration of the target in the original sample is calculated using the Poisson distribution: Target Concentration (copies/μL) = -ln(1 - p) (Total droplets / Volume per droplet), where p is the fraction of positive droplets.

Application: ddPCR excels in applications requiring ultra-sensitive and precise quantification of circular DNA variants. For mtDNA deletions, it has become the gold-standard for clinical heteroplasmy assessment due to its superior performance over qPCR. It can reliably detect and quantify heteroplasmy levels below 0.1%, which is crucial for identifying low-level mutations in asymptomatic carriers or in tissues where the mutation load is low (Qian et al., 2021). Its absolute quantification mitigates issues related to variable mtDNA copy number, as it can independently and absolutely count both mutant and wild-type molecules in the same reaction if using two differently colored probes. This allows for a direct calculation: % Heteroplasmy = [Mutant copies / (Mutant copies + Wild-type copies)] 100. For ecDNA monitoring in oncology, ddPCR's resilience to PCR inhibitors found in plasma makes it ideal for analyzing cell-free DNA from liquid biopsies. Specific probe sets can be designed for the unique junction of a patient's ecDNA, enabling ultrasensitive monitoring of minimal residual disease or emergence of therapy-resistant clones carrying the ecDNA, with a sensitivity far surpassing imaging or conventional sequencing for low-abundance targets.

Limitations: Despite its power, ddPCR shares a key constraint with qPCR: it is a targeted technique requiring precise prior knowledge of the sequence alteration to design mutation-specific probes. It cannot discover novel deletions or duplications. The upfront cost of instrumentation is higher than for qPCR, and the workflow, while highly automated, involves an extra partitioning step. The absolute quantification is also dependent on accurate droplet counting and assumes perfect partitioning, which, while highly reliable, can have technical margins of error. Finally, the multiplexing capacity (number of different targets measured in one well) is more limited compared to some advanced qPCR systems.

In the clinical pipeline, these methods often operate in tandem with broader discovery tools. A novel deletion might be identified in a patient via long-read sequencing of a muscle biopsy. Subsequently, a specific ddPCR assay can be developed for that unique breakpoint, enabling inexpensive, rapid, and ultrasensitive tracking of the mutation load in serial blood samples for non-invasive monitoring. Thus, qPCR and ddPCR translate the complex structural biology of circular DNA into actionable, numerical data for patient management, forming an essential bridge between research and the clinic.

Comparative Analysis and Recommendations for Method Selection

The diverse methodologies for detecting circular DNA length alterations each occupy a specific niche defined by resolution, throughput, cost, and required expertise. Selecting the optimal approach—or, more commonly, a synergistic combination—requires careful consideration of the biological question, sample type, and available resources. This section provides a comparative framework and practical recommendations to guide experimental design and clinical diagnostic strategy.

Comparative Summary of Methods

Table 1 presents a synthesized overview of the key technical and practical parameters for the major methods discussed.

Table 1. Comparative Analysis of Methods for Detecting Circular DNA Length Alterations.

Method	Resolution (Min. Detectable Change)	Throughput	Quantification	Relative Cost	Primary Application
Long-Range PCR	~50-100 bp (gel-based)	Medium	Semi-quantitative	$	Screening for known large deletions/duplications.
Southern Blot	~100-500 bp	Low	Semi- to Quantitative	$$	Validation of large rearrangements; topology confirmation.
2D Gel Electrophoresis	~500 bp (size), topology	Very Low	Qualitative	$$	Fundamental research on topology and complex populations.
Electron Microscopy (TEM)	Single bp (visual), ~50 bp (meas.)	Very Low	Quantitative (with stats)	$$$$	Definitive validation of circularity, direct length measurement.

Fiber-FISH	~1-5 kbp (on fibers)	Low	Quantitative (on fibers)	$$$	Structural mapping of large amplicons, physical length measurement.
Short-Read NGS	Single bp (breakpoints)	Very High	Digital, quantitative	$$	Genome-wide discovery of breakpoints; copy number analysis.
Long-Read NGS	Single bp (breakpoints)	High	Digital, quantitative	$$$$	Complete analysis: de novo assembly, phasing, complex SV detection.
qPCR	Single bp (assay-specific)	High	Relative quantitative	$	Quantitative screening & monitoring of known targets.
ddPCR	Single bp (assay-specific)	Medium	Absolute quantitative	$$	Ultra-sensitive quantification & clinical monitoring of known targets.

Abbreviations: bp, base pair; kbp, kilobase pair; SV, structural variant.

The resolution column highlights a critical distinction: methods like ddPCR and NGS can identify single-nucleotide breakpoints, while electrophoretic and microscopic methods are limited by gel migration or physical measurement precision. Throughput separates high-capacity sequencing and PCR methods from low-throughput validation techniques like TEM and Southern blot. Quantification ranges from qualitative (2D-GE) to precise absolute counting (ddPCR). Cost encompasses reagents, labor, and capital equipment, with long-read sequencing and TEM being the most resource-intensive.

Strategic Recommendations for Method Selection

The choice of method should be driven by a clear experimental or diagnostic goal. The following decision pathways are recommended:

1. Primary Screening for Known mtDNA Deletions/Duplications:
For diagnostic labs investigating common mitochondrial disorders, the priority is speed, cost-effectiveness, and reliable quantification of heteroplasmy. Long-range PCR offers a rapid first pass. qPCR provides more robust quantification and is suitable for most clinical thresholds. However, for the highest sensitivity required in carrier screening or monitoring low-level heteroplasmy, ddPCR is the recommended gold standard due to its precision and inhibitor tolerance (Qian et al., 2021). This tiered approach balances efficiency with accuracy.

2. Validation and Orthogonal Confirmation:
Discoveries from screening or NGS require validation with a method based on a different physical principle. For large mtDNA deletions or confirmation of ecDNA circularity, Southern blotting remains a powerful, if cumbersome, orthodoxy. In the modern context, long-read sequencing (PacBio HiFi or accurate ONT reads) is increasingly serving as the ultimate validation tool, as it provides both sequence confirmation and structural evidence in one assay, effectively replacing Southern blot for many applications (Zhao et al., 2021). A long read spanning a novel junction is incontrovertible proof.

3. Discovery of Novel, Unknown Rearrangements:
When the goal is unbiased discovery—such as identifying novel deletion species in aging tissue or characterizing the full spectrum of ecDNA in a tumor—high-throughput sequencing is mandatory. Short-read NGS (whole-genome or mtDNA-enriched) is cost-effective for large cohorts and excels at identifying breakpoints from split-read signals. However, for de novo assembly, resolving complex repeats, and phasing multiple variants on a single circle, long-read NGS is unequivocally superior. It moves from detecting a rearrangement to fully defining the resultant circular molecule.

4. Structural Analysis of Large ecDNA and Amplicons:
Understanding the architecture of oncogenic circles—the order of genes, the nature of duplications, and the presence of internal rearrangements—requires a combination of methods. Fiber-FISH provides a direct visual map, physically measuring distances between genetic loci on stretched DNA fibers, ideal for confirming tandem duplications (Shoshani et al., 2021). Electron microscopy gives the definitive proof of circularity and can reveal unexpected branching or multimeric structures. Long-read sequencing delivers the complete base-pair-resolution sequence and structural annotation. These three techniques together provide complementary views: visual mapping, physical topology, and complete sequence.

5. Integrated Clinical Diagnostic Pathway:
A comprehensive clinical diagnostic strategy for circular DNA-related diseases, particularly mitochondrial disorders, should be multi-stage. An effective pipeline could involve:

- Stage 1 (Screening): Use a targeted ddPCR panel for the most common pathogenic mtDNA mutations (e.g., common deletion, m.3243A>G) due to its speed, sensitivity, and quantitative precision for known variants.
- Stage 2 (Comprehensive Analysis): For screen-negative cases with high clinical suspicion, employ long-read mtDNA sequencing (either as a standalone assay or as part of a larger genetic panel). This captures the entire mitochondrial genome, detecting novel deletions, duplications, and point mutations simultaneously, and can assess heteroplasmy levels digitally from read counts (Stenton et al., 2021).
- Stage 3 (Monitoring): Once a pathogenic variant is identified, patient-specific ddPCR assays can be deployed for lifelong, non-invasive monitoring of heteroplasmy in blood or urine, assessing disease progression or response to therapy.

In conclusion, no single method is universally optimal. The researcher's or clinician's arsenal should include complementary techniques. The trend is clear: while foundational methods like PCR and Southern blot retain specific utilities, the future lies in the convergence of ultra-sensitive digital quantification (ddPCR) for monitoring and ultra-long molecular reads (long-read sequencing) for discovery and definitive characterization. This integrated approach will be crucial for unraveling the complex role of circular DNA length dynamics in human disease and translating these insights into improved diagnostics and therapies.

Conclusion and Future Directions

The detection and characterization of pathological alterations in the length of circular DNA represent a critical frontier in molecular diagnostics and disease mechanism research. As detailed in this review, the methodological arsenal available to researchers and clinicians is both extensive and evolving, ranging from foundational biochemical techniques to cutting-edge sequencing and imaging technologies. The optimal approach is never universal but is meticulously dictated by the specific biological question—be it screening for a common mitochondrial deletion, validating a novel oncogenic ecDNA structure, or discovering the complete circular genome landscape of a tumor. Key decision factors include the required resolution (from kilobase-scale changes to single-nucleotide breakpoints), necessary throughput, demand for absolute quantification, and available resources. The strategic integration of complementary methods—such as using long-read sequencing for discovery followed by ddPCR for ultrasensitive monitoring—often provides the most robust and clinically actionable insights.

Looking forward, several promising directions will shape the next generation of circular DNA analysis, moving beyond mere detection towards a systems-level understanding of its functional impact.

1. Integration of Multi-Omics Data. The future lies in correlating the physical structure and length of circular DNA with its functional outputs. Simply knowing a circle is elongated is insufficient; understanding its transcriptional and epigenetic activity is paramount. Future workflows will integrate long-read sequencing (for length and sequence) with companion assays like RNA-seq from

the same sample to link specific ecDNA structures to massive oncogene overexpression (Hung et al., 2021). Furthermore, leveraging the native epigenomic detection capability of Nanopore sequencing will allow researchers to simultaneously map the methylation landscape of a circular molecule, providing clues about its origin and regulatory state (Liu et al., 2021). This multi-modal integration will be essential for distinguishing passenger from driver alterations and for understanding how circular DNA structure directly influences gene regulation.

2. Advancement of Bioinformatics Tools. The power of long-read sequencing is currently bottlenecked by bioinformatic challenges. There is a pressing need for more sophisticated, dedicated algorithms for the de novo assembly, phasing, and structural variant calling specifically from circular DNA sequences. Current tools are often adapted from linear genome pipelines and struggle with the idiosyncrasies of circles, such as the lack of defined ends and the presence of tandem duplications. Next-generation algorithms must accurately resolve complex circular topologies, differentiate between single circles and concatemers, and confidently phase mutations on individual circular molecules to understand compound heteroplasmy in mtDNA or subclonal ecDNA evolution in cancer.

3. Single-Cell Analysis of Circular DNA. Cellular heterogeneity is a defining feature of both mitochondrial populations and tumors. Bulk analysis obscures the presence of rare cells harboring pathogenic mtDNA deletions at high heteroplasmy or aggressive subclones possessing oncogenic ecDNA. Emerging single-cell multi-omics technologies hold immense promise. Adaptations of single-cell ATAC-seq (scATAC-seq) or new methods for co-assaying DNA structure and transcriptomics in the same cell could, in theory, identify which specific cells carry shortened or elongated circles and simultaneously read out the consequent transcriptional program. For cancer, this would reveal the direct contribution of specific ecDNA structures to intra-tumoral heterogeneity, drug resistance, and metastatic potential at unprecedented resolution.

4. Exploration of Non-Coding Circular DNA (eccDNA). While focus has been on mtDNA and gene-amplifying ecDNA, the vast universe of small, non-coding extrachromosomal circular DNAs (eccDNAs) remains largely terra incognita. These molecules, often derived from repetitive or intergenic regions, are poorly understood but are ubiquitously present in cells, with their repertoire changing with age and disease. Developing specialized wet-lab and computational protocols to isolate, sequence, and—crucially—determine the length distribution of these eccDNAs is a major frontier. Understanding whether their length is regulated and whether specific size classes have functional roles (e.g., in innate immune signaling or gene regulation) could open an entirely new dimension of genome biology with potential diagnostic implications.

In conclusion, the accurate detection of circular DNA length alterations has transcended its status as a niche research endeavor. It is rapidly becoming an indispensable component of precision medicine. In mitochondrial diseases, it enables precise diagnosis, carrier screening, and therapeutic monitoring. In oncology, it provides insights into tumor evolution, mechanisms of therapy resistance, and offers novel biomarkers for liquid biopsy-based detection and monitoring. As methodologies continue to converge towards higher resolution, greater sensitivity, and single-cell applicability, our ability to decipher the code written in the circle will undoubtedly lead to transformative advances in diagnosing and treating a wide spectrum of human diseases.

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